

Tugas 5

Melanjutkan main.py untuk visualisasi data

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Kode lengkap `main.py`

```
import streamlit as st
import pandas as pd
from datetime import datetime
from config import *

# KONFIGURASI HALAMAN
st.set_page_config("Dashboard", page_icon="📊", layout="wide")

# DATA PELANGGAN
result_customers = view_customers()
df_customers = pd.DataFrame(result_customers, columns=[
    "customer_id", "name", "email", "phone", "address", "birthdate"
])
df_customers['birthdate'] = pd.to_datetime(df_customers['birthdate'])
df_customers['Age'] = (datetime.now() - df_customers['birthdate']).dt.days // 365

def tabelCustomers_dan_export():
    st.title("📄 Data Pelanggan")
    total_customers = df_customers.shape[0]
    col1, col2, col3 = st.columns(3)
    with col1:
        st.metric("📦 Total Pelanggan", total_customers, delta="Semua Data")

    st.sidebar.header("Filter Rentang Usia Pelanggan")
    min_age = int(df_customers['Age'].min())
    max_age = int(df_customers['Age'].max())
    age_range = st.sidebar.slider(
        "Pilih Rentang Usia",
        min_value=min_age,
        max_value=max_age,
        value=(min_age, max_age)
    )

    filtered_df = df_customers[df_customers['Age'].between(*age_range)]

    st.markdown("### 📄 Tabel Data Pelanggan")
    showdata = st.multiselect(
        "Pilih Kolom Pelanggan yang Ditampilkan",
```

```

        options=filtered_df.columns,
        default=list(filtered_df.columns)
    )
    st.dataframe(filtered_df[showdata], use_container_width=True)

    @st.cache_data
    def convert_df_to_csv(_df):
        return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

    csv = convert_df_to_csv(filtered_df[showdata])
    st.download_button("⬇️ Download Data Pelanggan sebagai CSV", data=csv,
file_name='data_pelanggan.csv', mime='text/csv')

# DATA PRODUCTS
result_products = view_products()
df_products = pd.DataFrame(result_products, columns=[
    "product_id", "name", "description", "price", "stock"
])

def tabelProducts_dan_export():
    st.title("📋 Data Produk")
    total_products = df_products.shape[0]
    total_stock = df_products['stock'].sum()
    total_value = (df_products['price'] * df_products['stock']).sum()

    col1, col2, col3 = st.columns(3)
    with col1:
        st.metric("📦 Total Produk", total_products)
    with col2:
        st.metric("📦 Total Stock", total_stock)
    with col3:
        st.metric("💰 Total Nilai Stock", f"Rp {total_value:,.0f}")

    showdata = st.multiselect(
        "Pilih Kolom Produk yang Ditampilkan",
        options=df_products.columns,
        default=list(df_products.columns)
    )

    st.markdown("### 📋 Tabel Data Produk")
    st.dataframe(df_products[showdata], use_container_width=True)

    @st.cache_data
    def convert_df_to_csv(_df):
        return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

    csv = convert_df_to_csv(df_products[showdata])
    st.download_button("⬇️ Download Data Produk sebagai CSV", data=csv,
file_name='data_products.csv', mime='text/csv')

```

```

# DATA ORDERS
result_orders = view_orders_with_customers()
df_orders = pd.DataFrame(result_orders, columns=[
    "order_id", "order_date", "total_amount", "customer_name", "phone"
])
df_orders['order_date'] = pd.to_datetime(df_orders['order_date'])
df_orders['order_date_only'] = df_orders['order_date'].dt.date

def tabelOrders_dan_export():
    st.title("📄 Data Orders")
    total_orders = df_orders.shape[0]
    total_revenue = df_orders['total_amount'].sum()

    col1, col2 = st.columns(2)
    with col1:
        st.metric("📦 Total Orders", total_orders)
    with col2:
        st.metric("💰 Total Revenue", f"Rp {total_revenue:,.0f}")

    st.sidebar.header("Filter Orders Berdasarkan Tanggal")
    start_date = st.sidebar.date_input("Mulai dari",
df_orders['order_date_only'].min(), key="start_order")
    end_date = st.sidebar.date_input("Sampai dengan",
df_orders['order_date_only'].max(), key="end_order")

    filtered_df = df_orders[
        (df_orders['order_date_only'] >= start_date) &
        (df_orders['order_date_only'] <= end_date)
    ]

    showdata = st.multiselect(
        "Pilih Kolom Orders yang Ditampilkan",
        options=filtered_df.columns,
        default=list(filtered_df.columns)
    )

    st.markdown("#### 📄 Tabel Data Orders")
    st.dataframe(filtered_df[showdata], use_container_width=True)

    @st.cache_data
    def convert_df_to_csv(_df):
        return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

    csv = convert_df_to_csv(filtered_df[showdata])
    st.download_button("⬇️ Download Data Orders sebagai CSV", data=csv,
file_name='data_orders.csv', mime='text/csv')

# DATA ORDER DETAILS
result_order_details = view_order_details_with_info()
df_order_details = pd.DataFrame(result_order_details, columns=[
    "order_detail_id", "order_id", "order_date", "customer_id", "customer_name",
    "product_id", "product_name", "unit_price", "quantity", "subtotal",

```

```

"order_total", "phone"
])
df_order_details['order_date'] = pd.to_datetime(df_order_details['order_date'])
df_order_details['order_date_only'] = df_order_details['order_date'].dt.date

def tabelOrderDetails_dan_export():
    st.title("📄 Data Orders Details")
    total_details = df_order_details.shape[0]
    total_revenue = df_order_details['subtotal'].sum()

    col1, col2 = st.columns(2)
    with col1:
        st.metric("📦 Total Order Items", total_details)
    with col2:
        st.metric("💰 Total Revenue (Detail)", f"Rp {total_revenue:,.0f}")

    st.sidebar.header("Filter Order Details")

    start_date = st.sidebar.date_input("Mulai dari",
df_order_details['order_date_only'].min(), key="start_od")
    end_date = st.sidebar.date_input("Sampai dengan",
df_order_details['order_date_only'].max(), key="end_od")

    customers_list = df_order_details['customer_name'].unique().tolist()
    selected_customers = st.sidebar.multiselect("Pilih Customer",
options=customers_list, default=customers_list)

    products_list = df_order_details['product_name'].unique().tolist()
    selected_products = st.sidebar.multiselect("Pilih Produk",
options=products_list, default=products_list)

    filtered_df = df_order_details[
        (df_order_details['order_date_only'] >= start_date) &
        (df_order_details['order_date_only'] <= end_date) &
        (df_order_details['customer_name'].isin(selected_customers)) &
        (df_order_details['product_name'].isin(selected_products))
    ]

    st.markdown("### 📄 Tabel Order Details")
    showdata = st.multiselect(
        "Pilih Kolom Order Details yang Ditampilkan",
        options=filtered_df.columns,
        default=list(filtered_df.columns)
    )
    st.dataframe(filtered_df[showdata], use_container_width=True)

    @st.cache_data
    def convert_df_to_csv(_df):
        return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

    csv = convert_df_to_csv(filtered_df[showdata])
    st.download_button("⬇️ Download Data Order Details sebagai CSV", data=csv,
file_name='data_order_details.csv', mime='text/csv')

```

```
# SIDEBAR UNTUK MEMILIH TABEL
st.sidebar.header("Pilih Tabel:")

show_customers = st.sidebar.checkbox("Tampilkan Pelanggan")
show_products = st.sidebar.checkbox("Tampilkan Products")
show_orders = st.sidebar.checkbox("Tampilkan Orders")
show_order_details = st.sidebar.checkbox("Tampilkan Order Details")

# HALAMAN AWAL
if not any([show_customers, show_products, show_orders, show_order_details]):
    st.title("📄 Selamat Datang di Dashboard")
else:
    if show_customers:
        tabelCustomers_dan_export()
    if show_products:
        tabelProducts_dan_export()
    if show_orders:
        tabelOrders_dan_export()
    if show_order_details:
        tabelOrderDetails_dan_export()
```

2. Penjelasan Kode `main.py`

Kode Customers

```
# Ambil data pelanggan dari database
result_customers = view_customers()
df_customers = pd.DataFrame(result_customers, columns=[
    "customer_id", "name", "email", "phone", "address", "birthdate"
])
df_customers['birthdate'] = pd.to_datetime(df_customers['birthdate'])
df_customers['Age'] = (datetime.now() - df_customers['birthdate']).dt.days // 365
```

Kode ini memanggil fungsi `view_customers()` dari file `config.py` untuk mengambil data pelanggan dari database. Setiap hasil query kemudian diubah menjadi DataFrame Pandas (`df_customers`). Kolom `birthdate` dikonversi menjadi tipe `datetime` agar bisa dihitung usianya, dan kolom `Age` dihitung berdasarkan selisih tanggal lahir dan tanggal saat ini.

Kode Fungsi meampilkan data

```
def tabelCustomers_dan_export():
    st.title("📄 Data Pelanggan")
    total_customers = df_customers.shape[0]
```

```

col1, col2, col3 = st.columns(3)
with col1:
    st.metric("📦 Total Pelanggan", total_customers, delta="Semua Data")

st.sidebar.header("Filter Rentang Usia Pelanggan")
min_age = int(df_customers['Age'].min())
max_age = int(df_customers['Age'].max())
age_range = st.sidebar.slider(
    "Pilih Rentang Usia",
    min_value=min_age,
    max_value=max_age,
    value=(min_age, max_age)
)

filtered_df = df_customers[df_customers['Age'].between(*age_range)]

st.markdown("### 📄 Tabel Data Pelanggan")
showdata = st.multiselect(
    "Pilih Kolom Pelanggan yang Ditampilkan",
    options=filtered_df.columns,
    default=list(filtered_df.columns)
)
st.dataframe(filtered_df[showdata], use_container_width=True)

@st.cache_data
def convert_df_to_csv(_df):
    return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

csv = convert_df_to_csv(filtered_df[showdata])
st.download_button("⬇️ Download Data Pelanggan sebagai CSV", data=csv,
file_name='data_pelanggan.csv', mime='text/csv')

```

Penjelasan Kode Produk

```

result_products = view_products()
df_products = pd.DataFrame(result_products, columns=[
    "product_id", "name", "description", "price", "stock"
])

```

Memanggil `view_products()` untuk mengambil semua produk. Data disimpan di `DataFrame` (`df_products`) dengan kolom penting: nama, deskripsi, harga, dan stock.

Kode fungsi menampilkan data

```

def tabelProducts_dan_export():
    st.title("📄 Data Produk")
    st.markdown("### 📄 Tabel Data Produk")

```

```

showdata = st.multiselect(
    "Pilih Kolom Produk yang Ditampilkan",
    options=df_products.columns,
    default=list(df_products.columns)
)
st.dataframe(df_products[showdata], use_container_width=True)

@st.cache_data
def convert_df_to_csv(_df):
    return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

csv = convert_df_to_csv(df_products[showdata])
st.download_button("⬇️ Download Data Produk sebagai CSV", data=csv,
file_name='data_produk.csv', mime='text/csv')

```

Fungsi `tabelProducts_dan_export` menampilkan data produk secara interaktif di dashboard Streamlit. User dapat memilih kolom mana saja yang ingin ditampilkan agar tabel lebih ringkas dan fokus, melihat tabel yang responsif menyesuaikan lebar layar, serta mengunduh data yang ditampilkan ke file CSV yang kompatibel untuk dibuka di Excel.

Penjelasan Kode Order

```

result_orders = view_orders_with_customers()
df_orders = pd.DataFrame(result_orders, columns=[
    "order_id", "order_date", "total_amount", "customer_name", "phone"
])
df_orders['order_date'] = pd.to_datetime(df_orders['order_date'])
df_orders['order_date_only'] = df_orders['order_date'].dt.date

```

Memanggil `view_orders_with_customers()` untuk mengambil semua order beserta info pelanggan. Data disimpan di DataFrame (`df_orders`), kolom `order_date` dikonversi ke datetime, dan dibuat kolom `order_date_only` untuk filter tanggal tanpa waktu.

Kode fungsi menampilkan data order

```

def tabelOrders_dan_export():
    st.title("📄 Data Orders")
    st.markdown("### 📄 Tabel Data Orders")
    showdata = st.multiselect(
        "Pilih Kolom Orders yang Ditampilkan",
        options=df_orders.columns,
        default=list(df_orders.columns)
    )
    st.dataframe(df_orders[showdata], use_container_width=True)

```

```

@st.cache_data
def convert_df_to_csv(_df):
    return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

csv = convert_df_to_csv(df_orders[showdata])
st.download_button("⬇️ Download Data Orders sebagai CSV", data=csv,
file_name='data_orders.csv', mime='text/csv')

```

Fungsi `tabelOrders_dan_export` menampilkan data pesanan beserta nama customer dan nomor telepon secara interaktif. User bisa memilih kolom yang ingin ditampilkan, melihat tabel responsif, dan mengunduh data ke file CSV. Tanggal order dikonversi agar memudahkan filter berdasarkan waktu, sehingga analisis lebih mudah.

Penjelasan Kode Order Details

```

result_order_details = view_order_details_with_info()
df_order_details = pd.DataFrame(result_order_details, columns=[
    "order_detail_id", "order_id", "order_date", "customer_id", "customer_name",
    "product_id", "product_name", "unit_price", "quantity", "subtotal",
    "order_total", "phone"
])
df_order_details['order_date'] = pd.to_datetime(df_order_details['order_date'])
df_order_details['order_date_only'] = df_order_details['order_date'].dt.date

```

Memanggil `view_order_details_with_info()` untuk mengambil detail tiap order beserta info customer dan produk. Data disimpan di DataFrame (`df_order_details`), kolom `order_date` dikonversi, dibuat kolom tambahan `order_date_only` untuk filter tanggal.

Kode fungsi menampilkan data order details

```

def tabelOrderDetails_dan_export():
    st.title("📄 Data Orders Details")
    st.markdown("### 📄 Tabel Detail Orders")
    customer_filter = st.sidebar.multiselect("Filter Customer",
options=df_order_details['customer_name'].unique(),
default=df_order_details['customer_name'].unique())
    product_filter = st.sidebar.multiselect("Filter Product",
options=df_order_details['product_name'].unique(),
default=df_order_details['product_name'].unique())
    filtered_df =
df_order_details[(df_order_details['customer_name'].isin(customer_filter)) &
(df_order_details['product_name'].isin(product_filter))]

    showdata = st.multiselect(
        "Pilih Kolom Detail Orders yang Ditampilkan",
        options=filtered_df.columns,

```



```

        default=list(filtered_df.columns)
    )
    st.dataframe(filtered_df[showdata], use_container_width=True)

    @st.cache_data
    def convert_df_to_csv(_df):
        return _df.to_csv(index=False, sep=';', encoding='utf-8-sig').encode('utf-8-sig')

    csv = convert_df_to_csv(filtered_df[showdata])
    st.download_button("⬇️ Download Data Detail Orders CSV", data=csv,
        file_name='data_order_details.csv', mime='text/csv')

```

Fungsi `tabelOrderDetails_dan_export` menampilkan detail setiap order, termasuk nama customer, produk, harga satuan, jumlah, subtotal, dan total order. User dapat memfilter berdasarkan customer dan produk, memilih kolom yang ingin ditampilkan, melihat tabel responsif, dan mengunduh data ke file CSV untuk analisis lebih lanjut.

Kode tampilan awal dan sidebar

```

# SIDEBAR UNTUK MEMILIH TABEL
st.sidebar.header("Pilih Tabel:")

show_customers = st.sidebar.checkbox("Tampilkan Pelanggan")
show_products = st.sidebar.checkbox("Tampilkan Products")
show_orders = st.sidebar.checkbox("Tampilkan Orders")
show_order_details = st.sidebar.checkbox("Tampilkan Order Details")

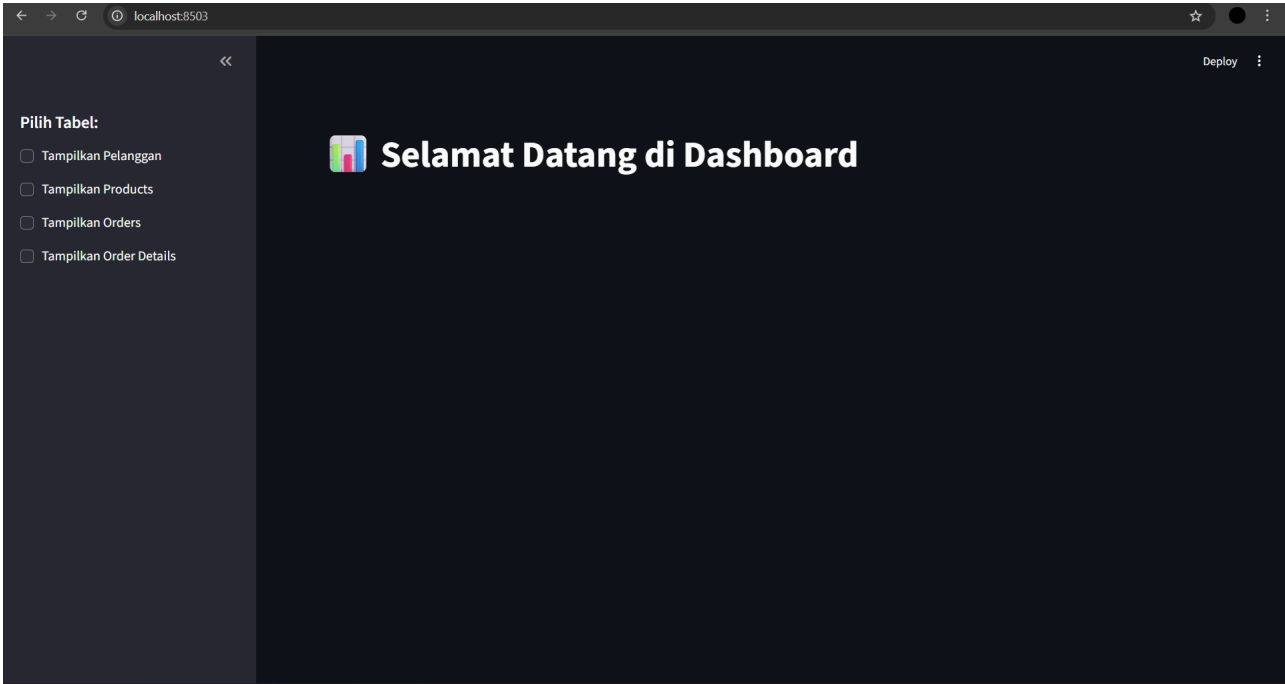
# HALAMAN AWAL
if not any([show_customers, show_products, show_orders, show_order_details]):
    st.title("🏠 Selamat Datang di Dashboard")
else:
    if show_customers:
        tabelCustomers_dan_export()
    if show_products:
        tabelProducts_dan_export()
    if show_orders:
        tabelOrders_dan_export()
    if show_order_details:
        tabelOrderDetails_dan_export()

```

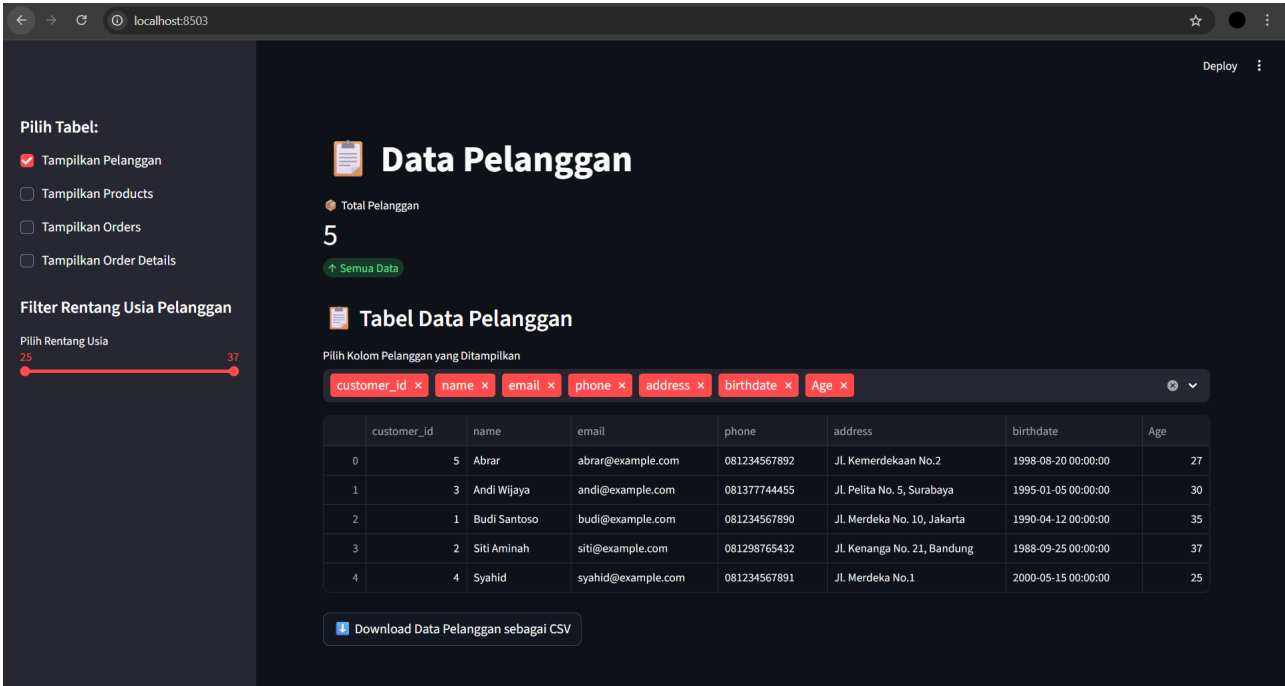
Kode ini mengatur navigasi utama dashboard menggunakan sidebar sebagai kontrol untuk memilih tabel mana yang ingin ditampilkan. Setiap checkbox mewakili satu jenis data: pelanggan, produk, orders, dan detail order. Jika tidak ada checkbox yang dipilih, aplikasi menampilkan halaman awal berupa judul dashboard.

Screenshot Tampilan

1. Tampilan halaman awal



2. Tampilan data pelanggan



3. Tampilan data produk

Pilih Tabel:

☐ Tampilkan Pelanggan

☒ Tampilkan Products

☐ Tampilkan Orders

☐ Tampilkan Order Details

Data Produk

Total Produk

5

Total Stock

125

Total Nilai Stock

Rp 146,500,000

Pilih Kolom Produk yang Ditampilkan

product_id xname xdescription xprice xstock x

Tabel Data Produk

	product_id	name	description	price	stock
0	5	Headset Logitech H390	Headset USB untuk PC/Laptop	300000	30
1	3	Keyboard Mechanical RK84	Keyboard mechanical hot-swap.	750000	20
2	1	Laptop Lenovo ThinkPad	Laptop untuk kebutuhan kerja dan mahasiswa.	8500000	10
3	4	Monitor Samsung 24"	Monitor 24 inch Full HD.	2000000	15
4	2	Mouse Logitech M331	Mouse wireless tanpa suara.	150000	50

Download Data Produk sebagai CSV

4. Tampilan data orders

Pilih Tabel:

☐ Tampilkan Pelanggan

☐ Tampilkan Products

☒ Tampilkan Orders

☐ Tampilkan Order Details

Filter Orders Berdasarkan Tanggal

Mulai dari

2025/11/20

Sampai dengan

2025/11/20

Data Orders

Total Orders

5

Total Revenue

Rp 12,650,000

Pilih Kolom Orders yang Ditampilkan

order_id xorder_date xtotal_amount xcustomer_name xphone xorder_date_only x

Tabel Data Orders

	order_id	order_date	total_amount	customer_name	phone	order_date_only
0	5	2025-11-20 16:21:50	450000	Abrar	081234567892	2025-11-20
1	4	2025-11-20 16:21:49	600000	Syahid	081234567891	2025-11-20
2	1	2025-11-20 12:56:53	8650000	Budi Santoso	081234567890	2025-11-20
3	2	2025-11-20 12:56:53	950000	Siti Aminah	081298765432	2025-11-20
4	3	2025-11-20 12:56:53	2000000	Andi Wijaya	081377744455	2025-11-20

Download Data Orders sebagai CSV

5. Tampilan data order details

←

→

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🌟

⌵

localhost:8503

Tampilkan Filter

Tampilkan Products

Tampilkan Orders

☒ Tampilkan Order Details

Filter Order Details

Mulai dari

2025/11/20

Sampai dengan

2025/11/20

Pilih Customer

Abbar

Syahid

Budi Santoso

Siti Aminah

Andi Wijaya

Pilih Produk

Mouse Logitech ...

Headset Logitech...

Laptop Lenovo T...

Keyboard Mecha...

📄

Data Orders Details

Total Order Items

8

Total Revenue (Detail)

Rp 12,600,000

📄

Tabel Order Details

Pilih Kolom Order Details yang Ditampilkan

order_detail_id

order_id

order_date

customer_id

customer_name

product_id

product_name

unit_price

quantity

subtotal

order_total

phone

order_date_only

	order_detail_id	order_id	order_date	customer_id	customer_name	product_id	product_name	unit_price	quantity	subtotal	order_total	phone	ord
0	8	5	2025-11-20 16:21:50	5	Abbar	2	Mouse Logitech M331	150000	1	150000	450000	081234567892	202
1	7	5	2025-11-20 16:21:50	5	Abbar	5	Headset Logitech H390	300000	1	300000	450000	081234567892	202
2	6	4	2025-11-20 16:21:49	4	Syahid	5	Headset Logitech H390	300000	2	600000	600000	081234567891	202
3	2	1	2025-11-20 12:56:53	1	Budi Santoso	2	Mouse Logitech M331	150000	1	150000	8650000	081234567890	202
4	4	2	2025-11-20 12:56:53	2	Siti Aminah	2	Mouse Logitech M331	150000	1	150000	950000	081298765432	202
5	1	1	2025-11-20 12:56:53	1	Budi Santoso	1	Laptop Lenovo ThinkPad	8500000	1	8500000	8650000	081234567890	202
6	3	2	2025-11-20 12:56:53	2	Siti Aminah	3	Keyboard Mechanical RK84	750000	1	750000	950000	081298765432	202
7	5	3	2025-11-20 12:56:53	3	Andi Wijaya	4	Monitor Samsung 24"	2000000	1	2000000	2000000	081377744455	202

📄

Download Data Order Details sebagai CSV

File lengkap ada di Github: <https://github.com/syahidrdho/Visualisasi-Data>

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